



Glass capsules of the sulfonamide Prontosil, 1936–1940

Fighting bacteria

- **What?** Sulfonamide drugs
- **Who?** Gerhard Domagk
- **Where and when?** Germany, 1932

Before antibiotics (bacteria-killing drugs) began to be widely used in the 1940s, another family of drugs—sulfonamides—was in use. Although certain sulfonamides

were effective against some bacterial infections, not all were safe, and one resulted in mass poisoning in 1937, causing more than 100 deaths in the US. This led the US government to introduce safety testing for new drugs for the first time in 1938.

Helping the heart

- **What?** Portable defibrillator
- **Who?** Frank Pantridge
- **Where and when?** UK, 1965

Defibrillators provide an electric shock to correct an abnormal heartbeat when someone is having a heart attack. Early machines were big and bulky, and they could be used only in hospitals.

Frank Pantridge's invention was small enough to be carried in ambulances. Today, many public places have portable Public Access Defibrillators (PADs), which can be operated by anyone.



Electrode pads are placed on the patient's chest.

Public Access Defibrillator, 2006

The device plays recorded instructions telling the user what to do, simply and calmly.

Lowering cholesterol

- **What?** Statin drugs
- **Who?** Akira Endo
- **Where and when?** Japan, 1971

Statins are drugs used to lower cholesterol, a fatty substance that can build up in arteries and block them, sometimes causing heart attacks. The drugs were developed after research into fungi, and have become some of the world's best-selling medicines.



Preventing malaria

- **What?** Artemisinin
- **Who?** Tu Youyou
- **Where and when?** China, 1972

The problem with many early antimalarial drugs, such as quinine, is that parasites eventually become immune to them. So, new drugs have to be invented—or old ones rediscovered. The Chinese chemist Tu Youyou produced artemisinin by retesting a 1,600-year-old treatment that used a plant called *Artemisia annua*. She was awarded the Nobel Prize in Medicine for her work.



Harvested *Artemisia annua* plants being prepared for sale in Uganda